

Methane-Air Sensitivity Analysis

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This note provides a simple sensitivity analysis of methane combustion in terms of oxidizer (restrained to air) temperature. A single 2-step mechanism (BFER) is considered. It represents methane partial oxidation and the subsequent carbon monoxide/dioxide equilibrium. Flow rates are dimensioned to match stoichiometric combustion to react a fixed volume flow rate of methane. Simulations are carried over a 2D axisymmetric domain with OpenFOAM 13 using the `multicomponentFluid` solver with a local time-stepping (LTS) advancement to establish steady-state solutions.

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Process conditions

In the present study we compare the combustion behavior of a fixed flow rate of methane (125 Nm³/h) under different combustion air temperatures (25 °C and 300 °C). Air flow rate is evaluated to match the global stoichiometric combustion of the supplied methane. All thermodynamics calculations (such as fuel heating value) and simulations (kinetics used in CFD) carried with BFER_methane mechanism from CERFACS. Table 1 summarizes process conditions and quantities of interest for verification.

Table 1: Summary of energy input conditions.

Property	Unit	Value
Required power	kW	1243.27
Lower heating value	MJ/kg	50.0254
Fuel mass flow rate	kg/h	89.4699
Oxidizer mass flow rate	kg/h	1532.35
Total mass flow rate	kg/h	1621.82
Fuel volume flow rate	Nm ³ /h	125
Oxidizer volume flow rate	Nm ³ /h	1190.48
Total volume flow rate	Nm ³ /h	1315.48
Water production	kg/h	200.935
Carbon dioxide production	kg/h	245.433
Total emissions	kg/h	446.368

Table 2 provides the main dimensions of the systems. Together with the values provided in Table 1 they are used to compute the boundary conditions that follow. These dimensions are used to simplify a more complex burner composed of 4 fuel injectors (of 27 mm each) while keeping surrounding air velocity constant by enforcing the ratio of cross-sections constant.

Table 2: System dimensions

Quantity	Unit	Value
Fuel inlet diameter	mm	27
Oxidizer inlet inner diameter	mm	32
Oxidizer inlet outer diameter	mm	199
Domain diameter	mm	866
Domain length	m	4

Table 3 provides the fuel operating conditions in units compatible with the simulation setup. These values are used for both cases considered in this study. Estimations of turbulent conditions assumed a turbulent intensity of 5%.

Table 3: Conditions evaluated for fuel.

Quantity	Unit	Value
Temperature	K	298.15
Mean velocity	m/s	66.1948
Characteristic length	mm	1.89
Turbulent kinetic energy	m^2/s^2	16.4316
Turbulent dissipation rate	m^2/s^3	5790.79

Table 4 provides room temperature air conditions computed under similar assumptions.

Table 4: Conditions evaluated for room temperature oxidizer.

Quantity	Unit	Value
Temperature	K	298.15
Mean velocity	m/s	16.5335
Characteristic length	mm	11.6707
Turbulent kinetic energy	m^2/s^2	1.02508
Turbulent dissipation rate	m^2/s^3	14.6125

Similarly, Table 5 provides the conditions for pre-heated air.

Table 5: Conditions evaluated for pre-heated oxidizer.

Quantity	Unit	Value
Temperature	K	573.15
Mean velocity	m/s	31.7832
Characteristic length	mm	11.6707
Turbulent kinetic energy	m^2/s^2	3.78814
Turbulent dissipation rate	m^2/s^3	103.806

Walls are modeled through a global heat transfer coefficient representing a refractory of 280 mm (thermal conductivity $1.8 \text{ W}/(\text{m.K})$) surrounded by steady air at room temperature (heat transfer coefficient $10 \text{ W}/(\text{m}^2.\text{K})$).

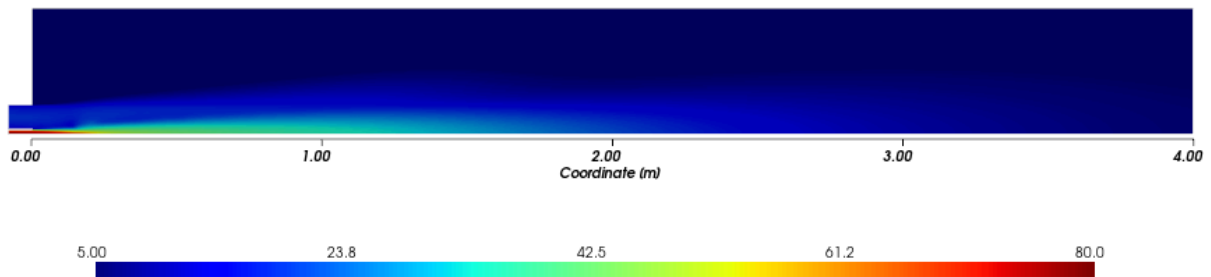
Results

In this section we present a selection of field views for the main variables of interest. All figures are structured the same way, with the reference room temperature results on top, and pre-heated air on the bottom. Care was taken to display both fields with the same color scale.

⚠ Warning

This is a work in progress, discussion of results is yet to be done.

Cold air inlet - Mean velocity (m/s)



Pre-heated air inlet - Mean velocity (m/s)

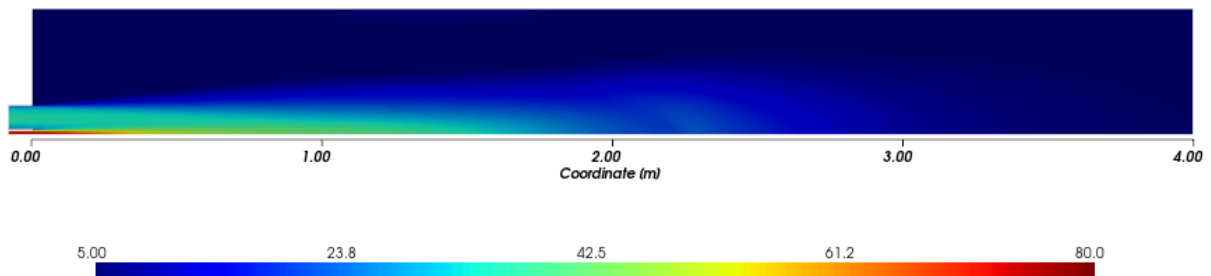
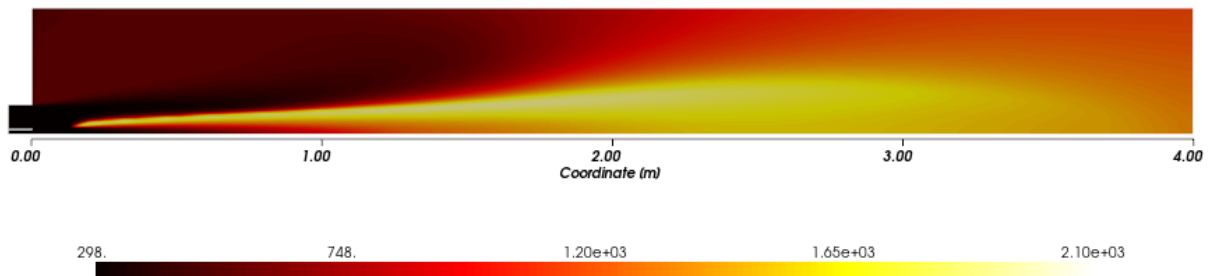


Figure 1: Velocity magnitude fields [m/s].

Cold air inlet - Temperature (K)



Pre-heated air inlet - Temperature (K)

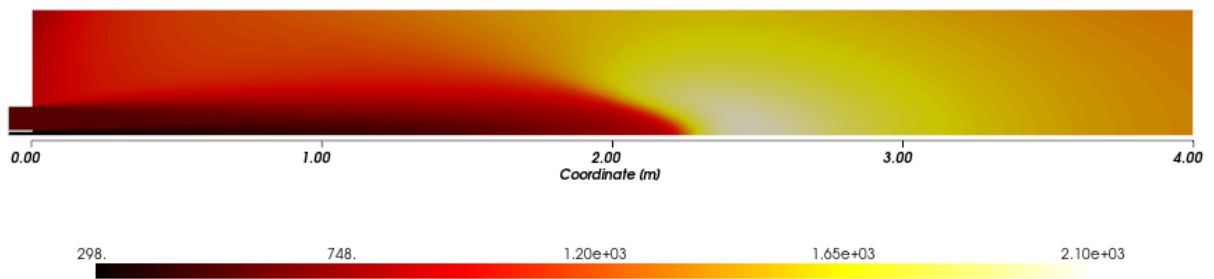
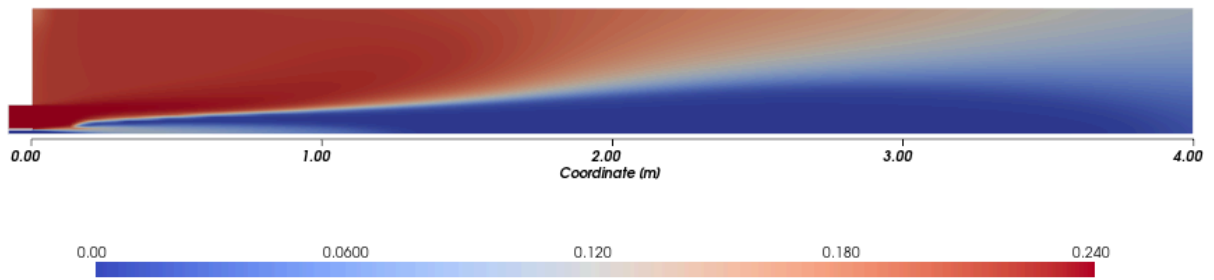


Figure 2: Temperature fields [K].

Cold air inlet - Oxygen mass fraction (-)



Pre-heated air inlet - Oxygen mass fraction (-)

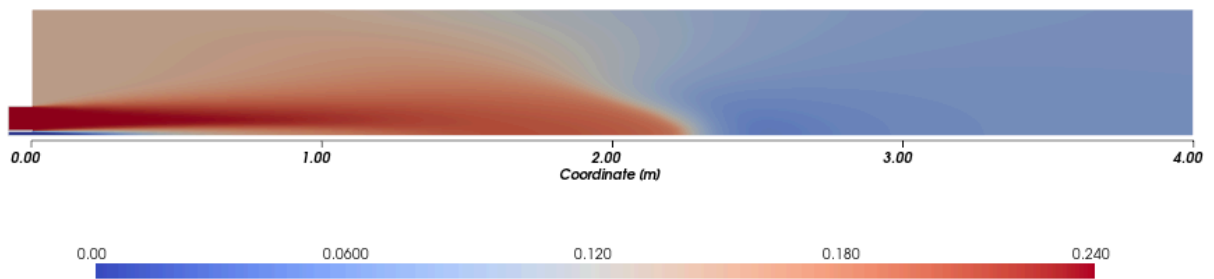
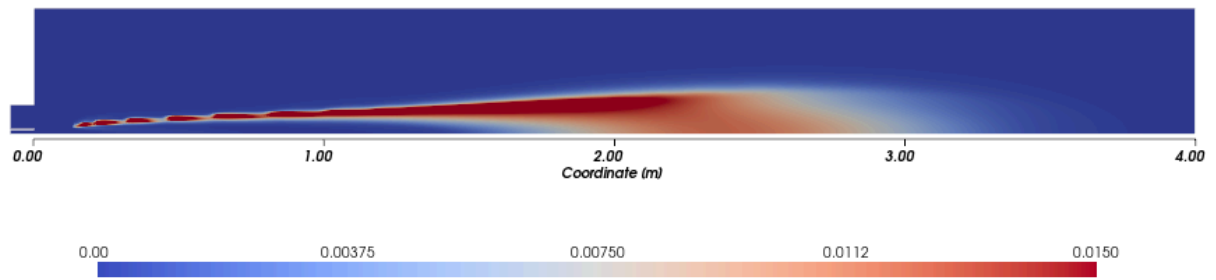


Figure 3: Oxygen mass fraction fields [-].

Cold air inlet - Carbon monoxide mass fraction (-)



Pre-heated air inlet - Carbon monoxide mass fraction (-)

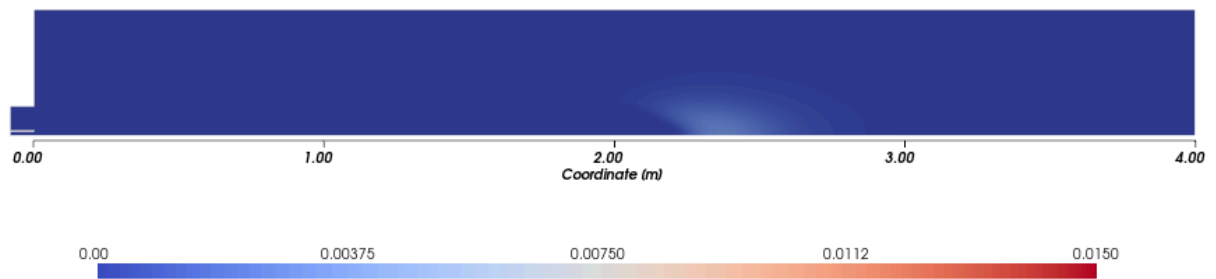
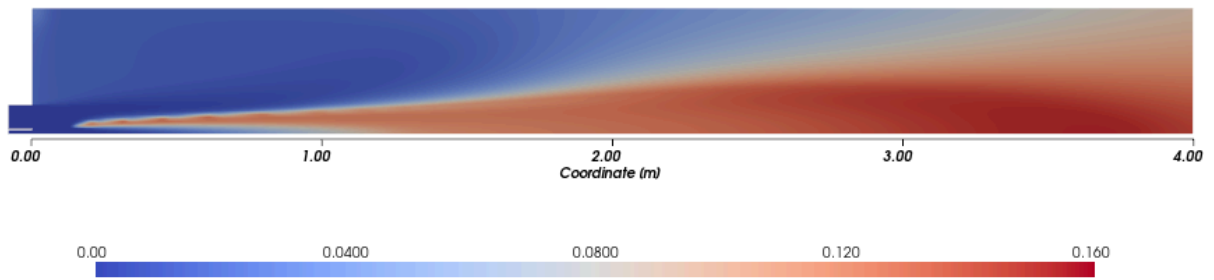


Figure 4: Carbon monoxide mass fraction fields [-].

Cold air inlet - Carbon dioxide mass fraction (-)



Pre-heated air inlet - Carbon dioxide mass fraction (-)

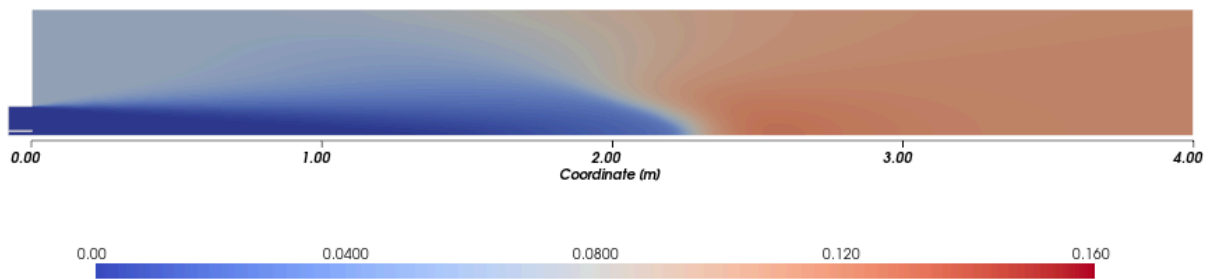
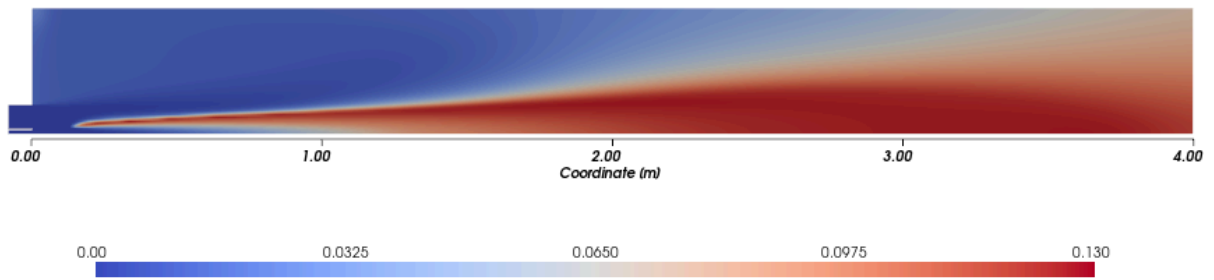


Figure 5: Carbon dioxide mass fraction fields [-].

Cold air inlet - Water vapor mass fraction (-)



Pre-heated air inlet - Water vapor mass fraction (-)

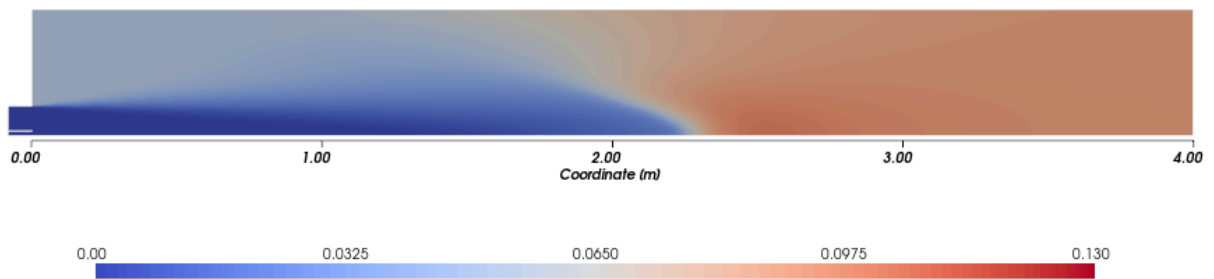
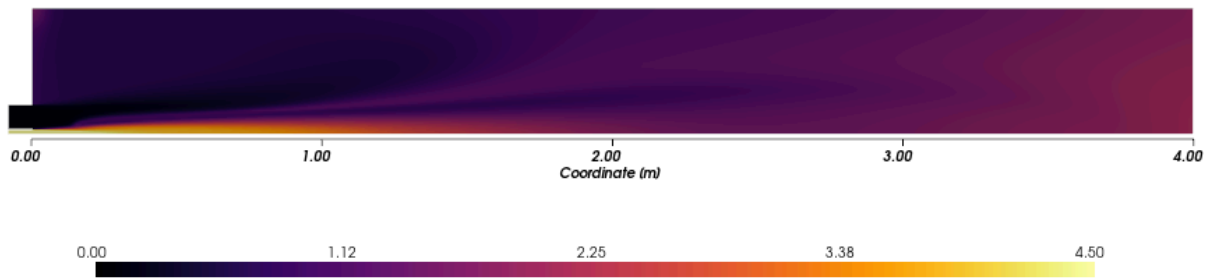


Figure 6: Water vapor mass fraction fields [-].

Cold air inlet - Net absorption coefficient (1/m)



Pre-heated air inlet - Net absorption coefficient (1/m)

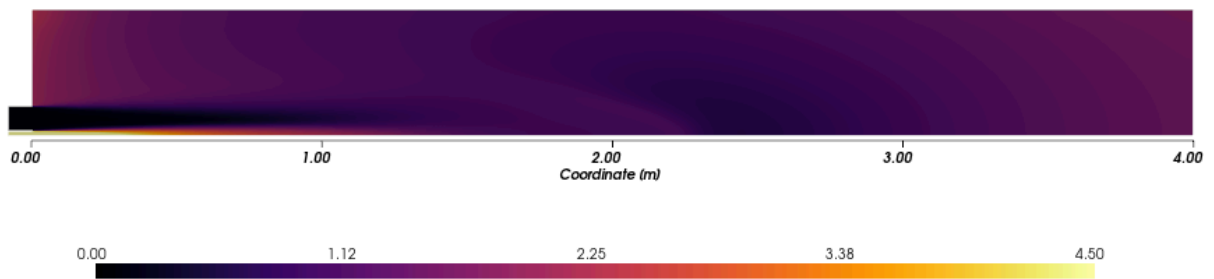


Figure 7: Net absorption coefficient fields [1/m].

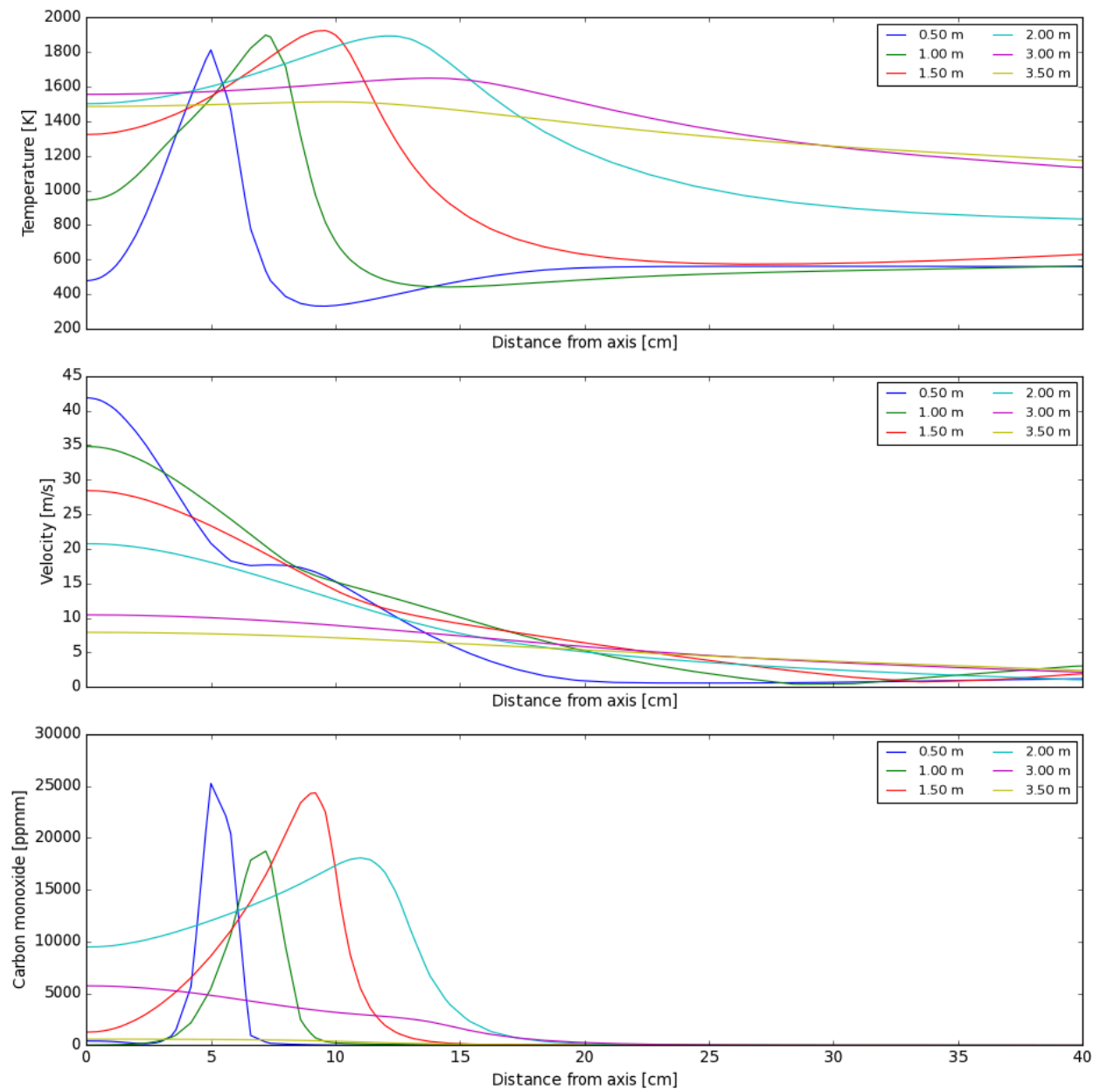


Figure 8: Extraction over selected axial coordinates (room temperature air).

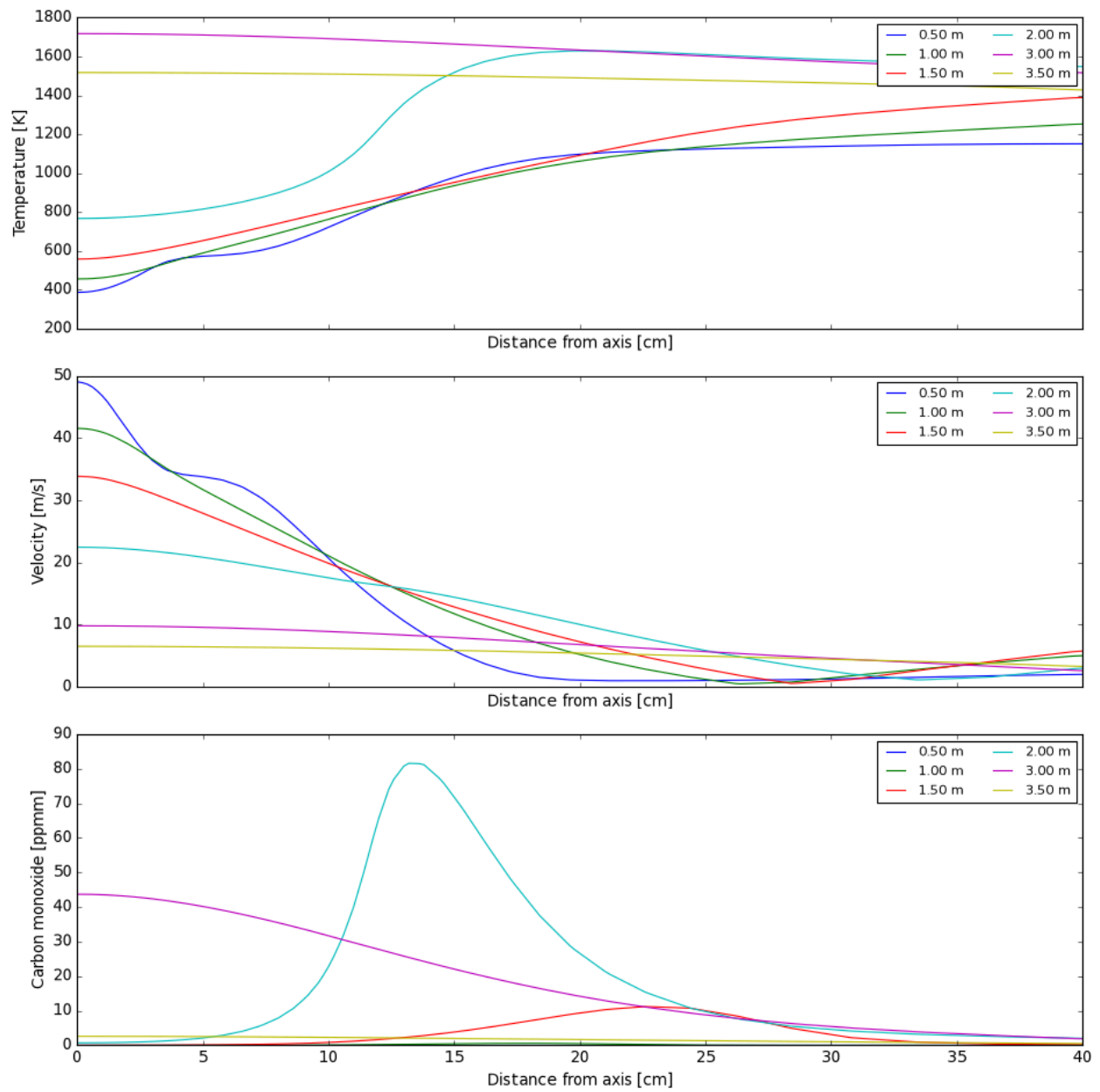


Figure 9: Extraction over selected axial coordinates (pre-heated air).